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Rapport Engineering Projects

Catégorie produit/service – Digital Trading Platform / AI Finance

Partenaire Extérieur : EFREI Paris – Pôle Léonard de Vinci

ArtEx

AI-Powered Fractional Art Trading & Perpetual Futures Platform

FP2ENG-25063-20260604

Authors: Yuhao Huang ; Yinghui Ren ; Lena Makri ; Berena-Jackie Douanla-Tchoutang-Malio ; Nelson-Harold Kamteu-Kamdem ; Dona-Prince-Ozias-Junior Mevi

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Résumé / Abstract

ArtEx is a proof-of-concept platform we built to explore what it would look like to combine digital art collecting with crypto-style derivatives trading. The core idea is that instead of paying millions for a single digital artwork, users can buy fractional shares of iconic NFT works and trade them like crypto assets — including opening leveraged perpetual contracts. To keep the platform usable before any real user base exists, we built a Virtual Market Maker (VMM) that simulates price movements using Geometric Brownian Motion. The frontend runs on Vue 3 with Tailwind CSS; the backend is planned as FastAPI with PostgreSQL, though it is not yet fully deployed. An AI chat assistant powered by DeepSeek helps users discover and evaluate artworks based on their preferences.

Revision History

Name / Version	Date	Changes
v0.1 – Initial scaffold	2025-10-01	Project kickoff; luxury art marketplace concept, Vue 3 frontend scaffold
v0.2 – AI Chatbot	2025-11-15	Integrated DeepSeek AI curator chatbot; API proxy pattern established

v0.3 – Crypto pivot	2026-01-20	Full redesign to crypto-native trading platform; VMM introduced; dark UI
v0.4 – Perpetual Futures	2026-02-10	Perpetual futures engine with leverage slider, TP/SL orders, liquidation logic
v0.5 – Wallet & Auth	2026-03-15	Mock auth (3 demo accounts), wallet connect (MetaMask, Phantom, Test Wallet)
v0.6 – AI Monitor	2026-04-20	AI Account Monitor ("小龍蝦接口"); 5-tab Portfolio view; Settings tab
v1.0 – Report Draft (25063)	2026-06-04	Final frontend feature-complete; backend scaffold planned; report generated

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Introduction

The NFT market hit USD 2.9 billion in 2021, then crashed — and we found that more interesting than discouraging. The crash exposed a real structural problem: most digital art platforms have no continuous liquidity mechanism, so when buyers disappear, prices simply stop moving. Before the crash, a single Beppele piece sold for USD 69 million in March 2021. The fact



that retail investors had almost no way to participate in that market — except by buying overpriced assets at peak hype — seemed like a genuine gap worth thinking about. That is what ArtEx is trying to address.

We started this as part of the EFREI Innovation Project in October 2025, without a completely fixed direction. The first idea was a luxury art marketplace. Somewhere around Sprint 3, after integrating live cryptocurrency prices and realising the dark UI looked better with a trading platform aesthetic, we shifted toward something more crypto-native. The combination of art aesthetics and leveraged financial instruments turned out to be more technically interesting — and more ethically complicated — than we originally expected.

This document covers what we built, why we made the technical decisions we did, and an honest account of where the project stands. The backend and real authentication are still at the planning stage, so what you will see in the demo is a complete frontend simulation.

Technical terms are defined in Section 3.2. The Gantt chart is in Appendix A and the API documentation is in Appendix B.

Documentation and Terminology

Reference Documents

Document	Number / URL	Attached?	Application
CoinGecko API v3 Documentation	https://docs.coingecko.com	No	Live cryptocurrency price feeds
DeepSeek API Reference	https://platform.deepseek.com/docs	No	AI chat completion (OpenAI-compatible)
Vue 3 Composition API Docs	https://vuejs.org/guide	No	Frontend framework specification
FastAPI Documentation	https://fastapi.tiangolo.com	No	Backend REST API framework
BlockCypher Testnet API	https://www.blockcypher.com/dev/bitcoin	No	Bitcoin testnet payment processing
Vite Build Tool Docs	https://vitejs.dev	No	Frontend build system & dev server
ArtEx Pitch Deck	ArtEx_Pitch_Deck.pptx (local)	Yes	Platform overview, business model, roadmap
Beeple — Everydays: The First 5000 Days	Wikimedia Commons	No	Source for primary artwork asset

Glossary

Terms

Term	Definition
Fractional Share	A unit representing proportional ownership of a tokenised artwork. Each artwork is divided into 1,000,000 shares.
Perpetual Future	A derivative contract with no expiry date that tracks the mark price of the underlying asset. P&L is settled continuously.
VMM (Virtual Market Maker)	A synthetic liquidity provider that generates bid/ask quotes and price history using a Geometric Brownian Motion model, ensuring tradability from day one.
Geometric Brownian Motion (GBM)	Stochastic process $S(t+1) = S(t) \times \exp((\mu - \sigma^2/2)\Delta t + \sigma\epsilon\sqrt{\Delta t})$ used to simulate asset price evolution.
Liquidation Price	The mark price at which a leveraged position is forcibly closed: $\text{liqPrice} = \text{entry} \times (1 \pm (1/\text{leverage} - 0.005))$.
TP/SL Order	Take-Profit / Stop-Loss conditional order that automatically executes when the mark price reaches a predefined level.
AI Account Monitor	An autonomous agent that periodically reads the user's portfolio state and submits structured action commands to an external AI endpoint.
Mark Price	The fair value price of an artwork share, computed by the VMM at each 2-second tick.
Collateral	BTC deposited as margin to open a leveraged perpetual position.

Acronyms

Acronym	Meaning	Explanation
NFT	Non-Fungible Token	Blockchain-based proof of ownership for a unique digital asset
VMM	Virtual Market Maker	Synthetic liquidity provider; see Terms above
GBM	Geometric Brownian Motion	Stochastic process for price simulation
API	Application Programming Interface	Standardised protocol for software communication
SPA	Single-Page Application	Web app where navigation occurs without full-page reloads
ORM	Object-Relational Mapper	SQLAlchemy layer translating Python objects to SQL queries
GDPR	General Data Protection Regulation	EU regulation on personal data processing
DeFi	Decentralised Finance	Financial services built on blockchain infrastructure
POC	Proof of Concept	Functional prototype demonstrating feasibility without production hardening
TRL	Technology Readiness Level	NASA/ESA scale 1-9 measuring maturity of a technology
TP/SL	Take-Profit / Stop-Loss	Conditional order types for automated risk management

Project Description – CDC – Lot 1 (3 pages)

Introduction and Strategic Challenges

Getting into the art market as a normal person is genuinely difficult. Prices are not public, attending the right auctions requires connections, and buying anything worth holding means spending tens of thousands of euros at minimum. The 2022 NFT crash made the digital side worse: platforms like OpenSea lost most of their volume because without a continuous mechanism to set prices, there is nothing stopping them from collapsing when buyers dry up. The main challenge for ArtEx was building something that works on day one — before any real user base exists — while still offering enough to attract traders who have other options.

We identified a few risks early that shaped our decisions. The biggest is regulatory: trading derivatives on art assets sits in a legal grey area, and we were never fully sure where the line between a trading platform and a gambling product falls under French law. Cold-start liquidity was the next concern — without our VMM, the platform would have no prices and nothing to trade. We also took API key security seriously; a lot of student projects leak keys into the browser bundle, which we specifically designed around. And AI hallucination is a genuine problem when an autonomous agent can execute trades based on LLM output.

Market Needs

Looking at existing platforms — SuperRare, OpenSea, Foundation — they all support buying and selling NFTs, but none of them offer leverage, fractional ownership, or any kind of AI advisory layer. On the finance side, Robinhood showed that retail investors will use simplified leveraged products if the interface makes it accessible. Artprice data puts the average NFT collector between 18 and 34 years old — the same group already using perpetuals on Binance. The gap we saw was that nobody had combined these two worlds in one product.

Product / Service Description

ArtEx delivers the following core features:

Fractional Share Trading: Each artwork is divided into 1,000,000 shares. Users buy/sell shares at the VMM mark price using cryptocurrency (BTC denominated).

Perpetual Futures: Users open long or short positions with 1–50× leverage, depositing BTC as collateral. Funding rates and liquidation mechanics mirror industry-standard CEX perpetuals (Binance, Bybit).

TP/SL Orders: Conditional orders that execute automatically when the mark price crosses a threshold; checked every 2 seconds.

AI Curator Chat: Embedded conversational AI (DeepSeek) recommends artworks based on user-expressed preferences.

AI Account Monitor: Autonomous agent that reads portfolio state and submits structured JSON actions (buy, sell, open/close perp, set TP/SL) to an external AI endpoint at configurable intervals.

Multi-Wallet Connect: Supports MetaMask, Coinbase Wallet, Phantom (Solana), and a Test Wallet for demo purposes.

Live Crypto Pricing: Top-5 cryptocurrencies (BTC, ETH, BNB, SOL, XRP) displayed in a ticker bar; all artwork prices denominated in user-selected crypto with fiat reference (6 currencies).

Innovative Nature of the Solution

What we think actually differentiates ArtEx from existing solutions:



VMM with GBM: Normal decentralised exchanges need real buyers and sellers on both sides. Our VMM generates synthetic prices using Geometric Brownian Motion, which means the platform is fully functional from the first second of deployment — even with zero real users. We found this the most technically satisfying part of the build.

Compounded volatility model: Artwork share prices move on their own via the VMM ($\sigma \approx 0.3/\text{year}$), while BTC moves independently in real time from CoinGecko ($\sigma \approx 0.8/\text{year}$). At 50× leverage, these two volatility sources compound in a way that does not exist anywhere in current financial markets. It is genuinely high-risk — a 1% BTC move can translate to a 50% P&L swing — which is also what makes it unusual as an asset class.

AI Account Monitor: This is the feature we are least certain about from a regulatory perspective. The monitor reads the user's portfolio state as JSON and sends it to an external LLM endpoint, which returns structured action commands that the platform then executes automatically. Machine-to-machine trading via LLM function-calling is not something we had seen done in an educational context before.

Freedom to Operate and Regulatory Constraints

Since ArtEx uses simulated transactions with no real money, none of these regulations apply to us right now. But when we thought about what a commercial version would actually face, the picture got complicated quickly.

MiFID II: Perpetual contracts on art assets would almost certainly count as complex financial instruments under EU law, requiring ESMA registration and retail suitability assessments before any real user could open a position.



GDPR: Trading history and AI interaction logs are personal data. A production platform would need EU-based data storage, explicit deletion mechanisms, and proper consent flows — none of which we have in the POC.

AML/KYC: Any actual crypto-to-fiat conversion would require identity verification under AMLD5/6. We did not implement this since there is no real money moving.

IP licensing: The artworks we used are from Wikimedia Commons under Creative Commons licences, which is fine for a student POC but would not hold up commercially. Real artist agreements would be needed.

Gambling law: This is the one that concerns us most. Leveraged speculation on near-randomly generated prices looks a lot like gambling under French law (Article L324-1 of the Code de la sécurité intérieure). We are not sure how that line is drawn in practice, and we suspect a specialist lawyer would not give a clean answer either.

Market Positioning and Competitive Advantage

SWOT Analysis:

Strengths	Weaknesses	
VMM guarantees liquidity at launch	Regulatory grey area (MiFID II / gambling law)	
Dual-asset volatility creates unique product	Simulated finance: no real monetary value in POC	
AI automation differentiates from manual platforms	Cold-start user acquisition challenge	
Low infrastructure cost (Vue SPA + FastAPI)	Artwork IP licensing not yet secured	

Opportunities	Threats

Growing retail DeFi user base (18–34 demographic)	Regulatory crackdown on crypto derivatives
AI agent investment tools gaining mainstream traction	Competitor platforms (Blur, SuperRare) with network effects
Art market digital transformation still early-stage	NFT market sentiment remains negative post-2022 crash
B2B white-label potential for art galleries	Smart contract audit costs if on-chain implementation pursued

Environmental Impact Assessment

Since ArtEx runs in the browser and on a local development server for the POC, the environmental footprint right now is essentially just our laptops. For a real deployment, the choices we made would keep the footprint manageable: cloud platforms like Vercel and Fly.io run on renewable energy, we store ownership in a database rather than on a blockchain (avoiding PoW energy costs entirely), and if we add on-chain settlement later, we would use Ethereum PoS which is roughly 99.95% more efficient than Bitcoin proof-of-work.

Cloud hosting (Vercel/Fly.io) uses renewable energy data centres, minimising Scope 2 emissions.

No blockchain transactions are executed in the POC — artwork ownership is maintained in a PostgreSQL database, avoiding PoW mining energy costs.

If real on-chain settlement is added, Ethereum PoS (post-Merge) reduces energy per transaction by ~99.95% vs. PoW Bitcoin.

Work Programme and TRL Assessment

Phase	Milestone	TRL	Status	
Phase 1	Frontend SPA: routing, design system, VMM, trading UI	TRL 3–4	COMPLETE	
Phase 2	AI integration: DeepSeek curator, AI Account Monitor	TRL 4	COMPLETE	
Phase 3	Backend: FastAPI, PostgreSQL, real auth, Alembic migrations	TRL 2–3	PLANNED	
Phase 4	Wallet integration (real EIP-1193), test-net BTC settlement	TRL 2	PLANNED	
Phase 5	Regulatory review, security audit, production deployment	TRL 5–6	NOT STARTED	

Function and Contribution of Actors – CDC – Lot 2 (1 page)

Team Organisation

The project team consists of EFREI ING3 students with the following responsibilities:

Actor	Role	Contribution
Yuhao Huang	Lead Developer & Architect	Frontend architecture (Vue 3), VMM design, AI integration, API proxy security pattern

Yinghui Ren	Frontend Developer	UI components, Tailwind CSS design system, responsive layouts, dark theme
Lena Makri	Product & UX	User experience design, competitive analysis, market research, pitch deck
Berena-Jackie Douanla-Tchoutang-Malio	Backend Developer	FastAPI scaffold, PostgreSQL schema, Alembic migrations (planned)
Nelson-Harold Kamteu-Kamdem	Blockchain & Wallet Integration	Wallet connect (EIP-1193), crypto payment flows, testnet BTC integration
Dona-Prince-Ozias-Junior Mevi	Data & AI	GBM parameter tuning, DeepSeek AI integration, AI Monitor design, regulatory research
EFREI Supervisor	Academic Supervisor	Project guidance, milestone validation, final evaluation

Team Motivation and Values

Honestly, we picked this project because it combined things we genuinely found interesting. Some team members trade crypto, some are interested in generative modelling, and one had been looking into AI trading agents independently. The innovation project required something technically ambitious, and the combination of stochastic price simulation, AI chat integration, and financial mechanics gave us more to actually build than the safer topics we considered. We were also drawn to the ethical tension: we knew from the start that what we were building looked similar to a gambling product, and we wanted to see if we could build it thoughtfully.

Project Management Methodology

We worked in roughly two-week sprints, though in practice the schedule was flexible — some sprints ran over when we hit unexpected problems. The biggest surprises were the crypto pivot in Sprint 3, which meant rebuilding most of the UI design from scratch, and the difficulty of making the AI Monitor reliable without a real backend. The team met online each week and used GitHub for version control and coordination.

Sprint	Period	Goal	Deliverable
1	Oct–Nov 2025	Concept & design	Wireframes, competitive analysis, tech stack decision
2	Nov–Dec 2025	Frontend scaffold & AI chat	Vue SPA + Tailwind, DeepSeek proxy, chatbot component
3	Jan 2026	Crypto pricing & dark redesign	CoinGecko integration, full dark UI, crypto denominated prices
4	Feb 2026	VMM & trading engine	GBM VMM, Pinia trading store, spot buy/sell
5	Mar 2026	Perpetual futures & wallet	Perp engine, leverage, TP/SL, wallet connect modal
6	Apr 2026	Auth, AI Monitor, Portfolio	Demo accounts, 5-tab AccountView, AI account monitor
7	May–Jun 2026	Polish, report & pitch deck	Logo, Settings, PowerPoint, this report

Technical Feasibility Studies – CDC – Lot 3 (3 pages)

Physical Environment

- The POC requires no dedicated physical infrastructure. Development and demo execution requires:
 - A development workstation running Windows 10/11, macOS 12+, or Ubuntu 22.04
 - A modern web browser (Chrome 120+, Firefox 122+, Edge 120+) with JavaScript enabled
 - Internet connectivity for CoinGecko price feeds and DeepSeek API calls
- For production deployment, a cloud-hosted VPS (e.g., Fly.io, Hetzner) with at least 2 vCPU / 4 GB RAM would be required to run the FastAPI backend and PostgreSQL database.

Hardware Environment

Component	Minimum Specification	Purpose
Development PC	Intel i5 / 8 GB RAM / SSD	Run Vite dev server + FastAPI locally
Browser	Chrome 120+ / Firefox 122+	Execute Vue SPA + crypto wallet extensions
Internet	5 Mbps stable connection	CoinGecko API polling (60s interval), DeepSeek AI calls
MetaMask Extension	Browser extension v11+	EIP-1193 wallet provider (optional — Test Wallet available)

Software Environment

Frontend dependencies (package.json):

Package	Version	Purpose
vue	3.x	Reactive UI framework (Composition API)
vite	5.x	Build tool, dev server, proxy configuration
tailwindcss	3.x	Utility-first CSS framework
pinia	2.x	State management (trading, prices, auth, wallet stores)
vue-router	4.x	Client-side routing (/ , /market, /trade/:id, /account, etc.)
axios	1.x	HTTP client for backend API calls

Backend dependencies (requirements.txt) — planned:

Package	Version	Purpose
fastapi	0.110+	Async REST API framework
sqlalchemy	2.x	Async ORM for PostgreSQL
alembic	1.x	Database migration management
pydantic	2.x	Request/response schema validation

psycopg2-binary	latest	PostgreSQL adapter
redis	latest	Caching layer for price data
python-jose	latest	JWT token auth
python-pptx	0.6+	PowerPoint generation (make_ppt.py)

External Services:

Service	Purpose	Authentication
CoinGecko API v3	Live BTC/ETH/BNB/SOL/XRP USD prices + 24h change	No API key (free tier, via Vite proxy /coingecko)
DeepSeek Chat API	AI art curator + account monitor LLM	Bearer token, injected server-side via Vite proxy /llm-api
BlockCypher Testnet	Bitcoin testnet payment processing	API token (checkout flow, /blockcypher proxy)
Wikimedia Commons	NFT artwork images (Beeple, Xcopy, Pak, Refik Anadol)	None (CC licensed)

Data

All state in the current POC is held in memory (Pinia stores) and partially persisted to localStorage. There is no active server-side database — that is planned for Phase 3. Here is how the data is structured:

Artwork data: 6 NFT works with metadata (title, artist, bio, tags, year, medium, edition, image URLs). Static in trading.js.

Price history: Ring buffer of last 60 VMM ticks per artwork (2-second resolution). ~480 data points in memory.

User wallet: {usd, btc, shares{}, perpPositions[], tpslOrders[], orderHistory[]}. Seeded from demo account on login.

Auth state: {username, displayName, color, initials}. Persisted in localStorage key artex_user.

Price preferences: {selectedCrypto, selectedFiat}. Persisted in localStorage keys artex_crypto, artex_fiat.

In a production backend, the following PostgreSQL tables would be required: users, artworks, share_holdings, perp_positions, orders, price_snapshots, ai_action_logs. GDPR compliance would require a data retention policy and anonymisation of trading logs after 90 days.

Test Suite – Specs (3 pages)

Tests are organised by functional requirement. Each requirement maps to a user story and validates one or more system behaviours. Tests are numbered globally (T01–T18). Nominal scenarios are presented first. The OK/NOK column is for jury validation during the live demo.

Requirement 1: Fractional Share Trading

Objective: A logged-in user with a connected wallet must be able to buy and sell fractional shares of any listed artwork at the current VMM mark price, with their portfolio updated in real time.

No.	Description	Execution Scenario	Expected Result	OK/NOK
T01	User logs in with demo account	Navigate to site; click person icon; select "Demo Trader"; click Quick Login	User is logged in; wallet seeded with \$10,000 USD and 0.1 BTC; portfolio tab accessible	
T02	Connect Test Wallet	Click "Connect Wallet"; select "Test Wallet"; click Connect	Wallet connected; address 0xTest4rtEx... displayed in header; green pill shown	
T03	Buy 100 shares at market price	Navigate to /trade/1 (Beeple); enter 100 in Buy Shares input; click Buy	Trade executes; BTC deducted from wallet; holdings updated; trade appears in order history	
T04	Sell 50 shares	In Holdings tab of AccountView; click Sell on Beeple; enter 50; confirm	BTC credited; share balance reduced by 50; order history updated	
T05	Insufficient funds rejection	Attempt to buy 10,000,000 shares (exceeds wallet balance)	Error message displayed; no trade executed; wallet unchanged	
T06	Price updates in real time	Observe mark price on any artwork card for 10 seconds	Price changes visibly every ~2 seconds; sparkline updates; MCap recomputes	

Requirement 2: Perpetual Futures Trading

Objective: A connected user must be able to open leveraged long/short positions, set TP/SL levels, monitor live PnL, and have positions liquidated automatically when margin falls below 0.5% of position value.

No.	Description	Execution Scenario	Expected Result	OK/NOK
T07	Open 10× long position	In TradingView Perp panel: select Long, set leverage 10×, deposit 0.005 BTC; click Open	Position created; displayed in Positions tab with entry price, size, liq price; BTC collateral deducted	
T08	Live PnL update	Observe open position card for 30 seconds	PnL value updates every ~2s; colour changes green/red with price movement	
T09	Set TP/SL on position	In Positions tab, click TP/SL on open position; enter TP +20% / SL -10%; save	TP/SL order created; visible in Orders tab; close price targets shown	
T10	Liquidation simulation	Open 50× short position with minimal collateral; wait for price to move against position	Position auto-closed at liquidation price; "Liquidated" entry in order history; toast notification shown	
T11	Close position manually	Click "Close" on an open position in Positions tab	Position closed at current mark price; PnL realised; BTC collateral + profit/loss returned to wallet	
T12	Open short position	Select Short, 5× leverage, 0.01 BTC collateral; open	Short position created with correct liq price above entry price; PnL positive when price falls	

Requirement 3: AI Curator Chat

Objective: The AI chat widget must call the DeepSeek API via Vite proxy, return a non-templated LLM response, and include relevant artwork recommendations with links.

No.	Description	Execution Scenario	Expected Result	OK/NOK
T13	AI responds to free-form preference	Open chat widget; type "I like dark abstract digital art with high volatility"; send	AI returns personalised response (not template); recommends ≥1 artwork; link to /trade/:id included	
T14	API key security	Open browser DevTools → Network; inspect POST to /llm-api	Authorization header NOT visible in browser; request headers show proxy forwarding only	
T15	Error graceful handling	Disconnect internet; send a chat message	Error message displayed ("Unable to reach AI service"); no crash; retry possible	

Requirement 4: AI Account Monitor

Objective: The AI Monitor must periodically send account state to an external AI endpoint, receive action commands, and execute them autonomously on the trading engine.

No.	Description	Execution Scenario	Expected Result	OK/NOK
T16	Configure and enable monitor	In AccountView → AI Monitor tab: enter endpoint URL, API key, set interval 30s; toggle Enable	Monitor activates; countdown timer displayed; status shows "Active"	
T17	Action command execution	Configure endpoint to return {"actions": [{"type": "buy_shares", "artworkId": 1, "shares": 10}]}	Monitor executes buy; action appears in Action Log with timestamp, type, and result	
T18	Action log persistence	After 3 monitor cycles; navigate away and return to AI Monitor tab	All 3 action entries remain visible in the log; no duplicate entries	

Analysis & Interpretation

Data Collection Protocols

Given the POC nature of the project, data collection was conducted through developer testing rather than instrumented user analytics. The following observations were collected during development:

VMM price series: 2-second ticks over 4-hour test sessions, producing ~7,200 price points per artwork. GBM parameters ($\mu=0.0001$, $\sigma=0.02$ per tick) were tuned to produce visually plausible volatility (~15% annualised).

API latency: DeepSeek API response time measured at 800ms–2.5s depending on prompt length. Vite proxy adds ~15ms overhead.

CoinGecko polling: Successful 60-second price refresh over 2-hour sessions without rate limiting on the free tier.

Results Presentation

Key functional results achieved at POC milestone (v1.0, June 2026):

Metric	Target	Achieved	Notes
VMM liquidity	Continuous, no gaps	Yes — 2s tick rate	6 artwork markets always have a mark price
AI chat latency	< 5 seconds	Avg 1.8s	DeepSeek API p95 ≈ 3.2s
Leverage range	1×–50×	1×–50× implemented	Slider with real-time liq price preview
Wallet options	≥ 3 providers	5 providers (incl. Test Wallet)	MetaMask, Coinbase, Phantom, WalletConnect, Test
Currency options	≥ 3 crypto + 3 fiat	5 crypto + 6 fiat	Persisted in localStorage
AI Monitor actions	All 6 action types	6 types implemented	buy, sell, open_perp, close_perp, set_tpsl, cancel_tpsl
Mobile responsive	Tablet + desktop	Partial — desktop optimised	Mobile breakpoints incomplete

Analysis Methods

To check whether our VMM produced believable prices, we compared a 30-day simulated series against historical Beeple NFT price data from OpenSea. The GBM model at $\sigma=0.45/\text{year}$ gives visually plausible results, but it consistently underestimates sharp drops — the kind of sudden 50–70% collapse that happened across NFT markets in 2022. A jump-diffusion model, like Merton's 1976 formulation, would handle this more accurately. We did not implement it because of time and complexity constraints, but we think it would be the right direction for Phase 4.

We tested the AI curator manually across around 25 different prompts. DeepSeek gave contextually reasonable responses in about 22 of them — the failures were mostly when the user input was too vague, like 'I like everything,' which gives the model nothing to work with. Roughly 88% accuracy is probably not high enough for a production product, but for a POC it showed the approach works.

Interpretation & Discussion

The VMM does what we needed it to do: prices never stop, and the platform is always tradeable. But we want to be clear about what that actually means. The prices are generated by an algorithm, not by real supply and demand meeting. Whether that makes it more like a prediction market, a casino, or something in between is a question we do not have a clean answer to. We addressed this directly in the pitch deck because we felt it was important to say out loud rather than bury.

The AI Account Monitor works technically, but the liability question is genuinely unresolved. If an AI agent executes a sequence of trades that causes a large loss — even in simulation — who is accountable? In a POC it does not matter. In a real product it would matter a great deal, and current regulation under MiFID II Article 17 (algorithmic trading accountability) is not specific enough to give a clear answer.

The dual-volatility model — art price plus BTC price multiplied through leverage — creates a risk exposure that does not exist in any current financial product. We found it interesting to build, but we also noticed it is very hard to explain clearly to a normal person. A retail investor at 50× leverage facing two independent sources of volatility may not be in a position where

informed consent is really meaningful.

Conclusions

The POC shows that all the core mechanics work. We built a platform that runs, generates prices, lets users open leveraged positions, and connects to a real AI API — without spending anything on infrastructure. The VMM solved the liquidity problem. The AI integration worked better than we expected. The frontend is complete and demo-ready.

The gap between this and a real product is mostly legal rather than technical. The mechanics of the platform — leverage combined with near-random price simulation — are difficult to distinguish from gambling. Any path to commercial operation would require a legal opinion and likely a fundamental product redesign. We think being direct about that is more useful than papering over it.

Projection & Roadmap

Phase	Timeline	Goal	KPI
Phase 3: Backend	Q3 2026	Real auth, PostgreSQL persistence, REST API	All data persists across sessions
Phase 4: On-chain	Q4 2026	Ethereum testnet settlement, real wallet transactions	First real ETH transaction on testnet
Phase 5: Legal	Q1 2027	Regulatory assessment, GDPR compliance, IP licences	Legal opinion from EU financial lawyer
Phase 6: Beta	Q2 2027	Closed beta with 100 real users, real small-value trades	100 registered users, 1,000 trades
Phase 7: Launch	Q4 2027	Public launch with ≥10 partner artists	10 artists, €100K trading volume

Environmental Impact & Eco-Design

Environmental impact was something we thought about seriously during the project — partly because it is a formal requirement, and partly because NFTs have a genuinely bad reputation for energy use. This section explains our thinking and what we actually did about it.

Digital Art vs. Physical Art: Environmental Baseline

Physical art has environmental costs that do not get talked about much. International air freight for paintings, climate-controlled warehouses, printed catalogues, and events like Art Basel Geneva — which reportedly generates around 30,000 tonnes of CO₂e per edition. ArtEx has none of those costs. There is nothing to ship, no warehouse needed, and no event required to reach buyers.

Impact Category	Traditional Art Market	ArtEx (Digital)	Reduction
Artwork transport	Air/road freight per sale	Zero — digital delivery	~100%

Physical print production	Paper, ink, framing per unit	Zero — no physical goods	~100%
Storage & climate control	Warehouse energy 24/7	Cloud storage only	>95%
Art fair attendance	Flights, hotels per event	Online platform — no events	~100%
Server infrastructure	N/A	Cloud compute (see below)	New footprint (minimised)

Platform Energy Consumption & Carbon Footprint Estimate

The platform's energy footprint comes from three things: server compute, users' devices, and the AI API calls for the chat features. For the current POC running on local machines, the footprint is essentially our laptops.

Source	Consumption Estimate	CO ₂ e Equivalent	Notes
<u>Vite dev server (local)</u>	<u>~15W × dev hours</u>	<u>Negligible (<0.1 kg/month)</u>	<u>POC only; no production server</u>
<u>End-user browser (SPA)</u>	<u>~5–10W per active session</u>	<u>~0.3 g CO₂e per page view</u>	<u>Industry average for SPA</u>
<u>CoinGecko API calls</u>	<u>60s polling × 5 cryptos</u>	<u>Shared server; negligible marginal cost</u>	<u>Free tier, no dedicated compute</u>
<u>DeepSeek AI inference</u>	<u>~1 API call per chat message</u>	<u>~0.2–0.5 g CO₂e per call</u>	<u>Shared GPU cluster; estimated</u>
<u>VMM computation (client)</u>	<u>JS GBM tick every 2s (browser)</u>	<u>~0.1W additional CPU load</u>	<u>Client-side; no server cost</u>

At around 1,000 daily active users in production, our rough estimate puts the total at 15–30 kg CO₂e per month. That is comparable to driving a car about 80–150 km, or an economy flight covering roughly 8 km of distance. Not zero, but dramatically lower than any equivalent physical art operation.

Blockchain & Cryptocurrency Considerations

Early on we decided not to use real blockchain transactions in the POC — partly for environmental reasons, partly because it would have added months of complexity. Bitcoin uses roughly 700–900 kWh per transaction (about a month of typical household electricity), so avoiding on-chain settlement removes the largest potential footprint item entirely. If we add real settlement in a later phase, we plan to use Ethereum PoS, which cut energy per transaction by about 99.95% compared to proof-of-work. Layer 2 networks like Polygon or Arbitrum would reduce it by roughly another 99% on top of that.

Bitcoin (PoW): A single Bitcoin transaction consumes approximately 700–900 kWh — equivalent to a month of household electricity. Using BTC only as a pricing unit (not for settlement) avoids this entirely.

Ethereum (PoS post-Merge): Since September 2022, Ethereum's Proof-of-Stake consensus reduces energy per transaction by ~99.95% vs. PoW. If on-chain settlement is added in Phase 4, Ethereum PoS is the mandated choice.

Layer 2 solutions: For production, integrating a Layer 2 network (Polygon, Arbitrum, Base) would reduce per-transaction energy further by ~99% vs. Ethereum L1, at roughly 0.0003 kWh per transaction.

In the current POC, the 'BTC' denomination is just a number in a JavaScript store. No real Bitcoin moves, so blockchain-related emissions are zero.

Eco-Design Principles Applied During Development

During development, we tried to make choices that kept the platform lean — which also happens to be good for performance and hosting cost.

Principle	Implementation	Environmental Benefit
Efficient data transfer	SPA architecture — only changed components re-render; no full-page reloads	Reduced network requests and CPU cycles per interaction
API call minimisation	CoinGecko polled every 60s (not real-time); results cached in Pinia store	Fewer server-side computations; lower API provider energy use
Client-side VMM	Price simulation runs in browser JS, not on a dedicated server	Eliminates continuous server compute for price generation
Static asset optimisation	Vite production build: tree-shaking, code splitting, minification	Smaller JS bundles → faster load → less energy per page view
Image efficiency	Artwork images loaded from Wikimedia Commons CDN (globally cached)	Leverages existing CDN cache; avoids redundant origin fetches
No blockchain settlement (POC)	Ownership stored in PostgreSQL, not on-chain	Zero PoW/PoS transaction energy in current implementation
Green hosting target	Production target: Vercel (Scope 2: 100% renewable) or Fly.io (RE100)	<u>Net-zero Scope 2 emissions for server infrastructure</u>

Lifecycle Assessment Summary

Thinking about the full lifecycle in rough terms:

Development (now): The main energy cost was our laptops and occasional DeepSeek API calls. We estimate somewhere around 50–100 kg CO₂e total for the whole project — roughly equivalent to a five-hour flight.

Production (planned): At 1,000 DAU, around 15–30 kg CO₂e per month, dominated by AI inference costs. Near-zero if we use green hosting for our own infrastructure.

Decommissioning: No physical hardware to dispose of. Shutting down cloud instances and deleting user data in line with GDPR Article 17 is all that is required.

Regulatory Compliance: Environmental Law

There are a few EU environmental regulations that would apply at commercial scale, though none of them currently affect us as a student POC.

EU Taxonomy Regulation (2020/852): Digital trading platforms are not currently in scope for what counts as 'environmentally sustainable' under the taxonomy, but the regulation is expanding and could become relevant.

CSRD: Large companies (>250 employees) must report Scope 1, 2, and 3 emissions. Not relevant at POC stage.

French Loi REEN (2021): Requires large digital platforms to publish annual digital sobriety reports — would apply if the platform reaches more than 100,000 French users.

MiCA (2023): Crypto-asset service providers must disclose their consensus mechanism's environmental impact. Since we do not run our own blockchain, this requirement would be minimal for ArtEx.

Commitments & Future Roadmap

For a future production version, our environmental commitments would be: deploying on renewable-energy hosting (Vercel and Fly.io both qualify), using only PoS or Layer 2 if we add on-chain settlement, running an annual review of API call frequency and bundle size, and showing users an estimated CO₂e footprint per trading session in their account settings.

Commitment	Target	Timeline
<u>100% renewable energy hosting</u>	<u>Deploy on Vercel or Fly.io (RE100 certified)</u>	Phase 5 (Q1 2027)
Carbon-neutral AI inference	Select AI provider with verified carbon offsets or use efficient local model	Phase 5 (Q1 2027)
Blockchain: PoS only	If on-chain settlement added, use Ethereum PoS or L2 exclusively	Phase 4 (Q4 2026)
Digital sobriety audit	Annual review of API call frequency, bundle size, and server compute	Annual from Phase 6
User transparency	Display estimated CO ₂ e footprint per trading session in account settings	Phase 6 (Q2 2027)
Artwork IP sustainability	Partner only with artists who produce work without physical waste	Phase 7 (Q4 2027)

Conclusions & Perspectives

We built a working digital art trading platform in one academic year, as a team of six students, with no infrastructure budget. That surprised us a little. The trading engine, the AI integration, the VMM, and the perpetuall all work. The frontend is complete and ready to demonstrate.

Commercially, the niche ArtEx targets is real — we did not find any other platform that combines leverage, AI automation, and digital art in the same product. Masterworks does fractional art without leverage. Polymarket does speculation without art. The first-mover window is open, but probably not for long as regulation around crypto derivatives tightens across the EU.

The main risks going forward are regulatory. There is a real chance that what we built would be classified as a gambling product under French law, which would require either an ARJEL licence or a significant redesign. The NFT market is still depressed from the 2022 crash, which makes user acquisition harder. And if the AI Monitor ever executes a bad trade sequence in a real product, the reputational damage could be serious even if the legal liability is ambiguous.

One decision we made early and kept: we did not implement features that would have increased engagement at the cost of user welfare. No push notifications designed to pull users back during price spikes. No 'this is trending' labels to manufacture urgency. No option to reload a liquidated position by paying more. The platform we built is probably not the most profitable version of this idea, and we made that trade-off deliberately.

As an academic project, ArtEx met its objectives. We built something technically real — not a wireframe or a mock, but a working application — that shows how full-stack web development, stochastic financial modelling, and AI API integration can fit together. We also came away with a clearer view of why the more financially interesting a product is, the more carefully it needs to be designed around who it might harm.

Appendix A — Project Gantt Chart

Simplified Gantt representation (full interactive version available in project repository):

Task	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
Competitive analysis & wireframes	■	■							
Vue 3 + Tailwind scaffold		■	■						
DeepSeek AI integration		■	■						
Dark UI redesign + crypto pricing				■					
VMM (GBM) implementation				■	■				
Perpetual futures engine					■	■			
Wallet connect (MetaMask/Test)						■			
Auth system + demo accounts							■		
AI Account Monitor							■		
Portfolio AccountView (5 tabs)							■	■	
Settings + Logout								■	
Pitch deck (PPT)								■	
Project Report (this document)									■

Appendix B — API Reference

Vite Proxy Routes (Frontend Dev Server)

Proxy Path	Target	Auth	Notes
/api/*	http://localhost:8000	None (local)	FastAPI backend
/llm-api/*	https://api.deepseek.com	Bearer injected server-side	LLM_API_KEY from .env (no VITE_ prefix)
/coingecko/*	https://api.coingecko.com	None	Free tier; polling 60s intervals
/blockcypher/*	https://api.blockcypher.com	Token query param	BTC testnet payments

AI Account Monitor — Action Command Schema

Request body sent to the external AI endpoint (POST application/json):

```
{ "wallet": { "usd": number, "btc": number, "shares": {...} }, "positions": [...], "tpsOrders": [...], "prices": {...}, "timestamp": ISO8601 }
```

Expected response schema:

`{ "actions": [ { "type": "buy_shares" | "sell_shares" | "open_perp" | "close_perp" | "set_tpsl" | "cancel_tpsl", ...params } ] }`

Action Type	Required Parameters	Description
buy_shares	artworkId: number, shares: number	Buy fractional shares at current mark price
sell_shares	artworkId: number, shares: number	Sell shares at current mark price
open_perp	artworkId: number, direction: "long" "short", leverage: number, btcCollateral: number	Open leveraged perpetual position
close_perp	positionId: string	Close open position at current mark price
set_tpsl	positionId: string, tp: number null, sl: number null	Set/update take-profit and stop-loss levels
cancel_tpsl	orderId: string	Cancel a pending TP/SL order